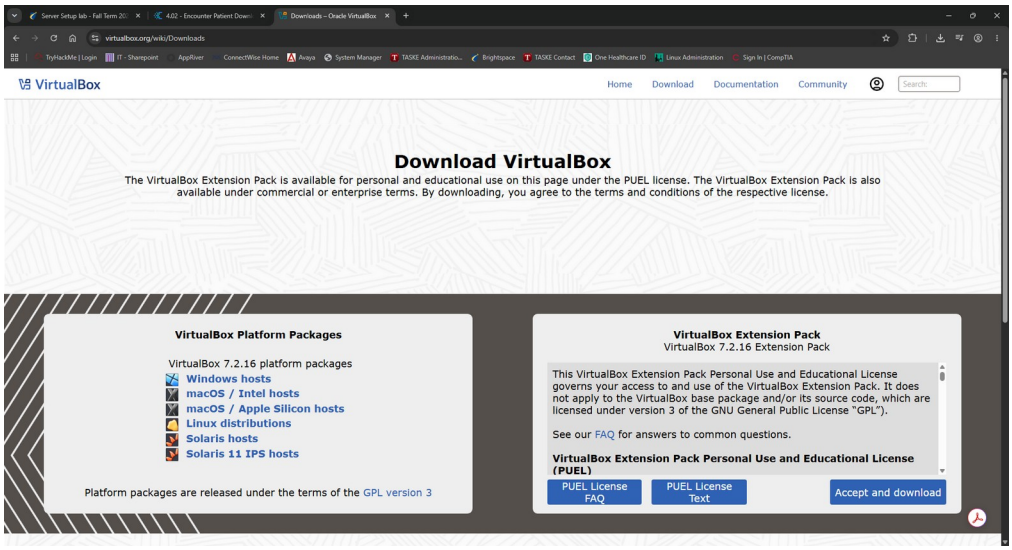
Joel Rodriguez
Fall 2026
CIS-245
Setup Server Lab

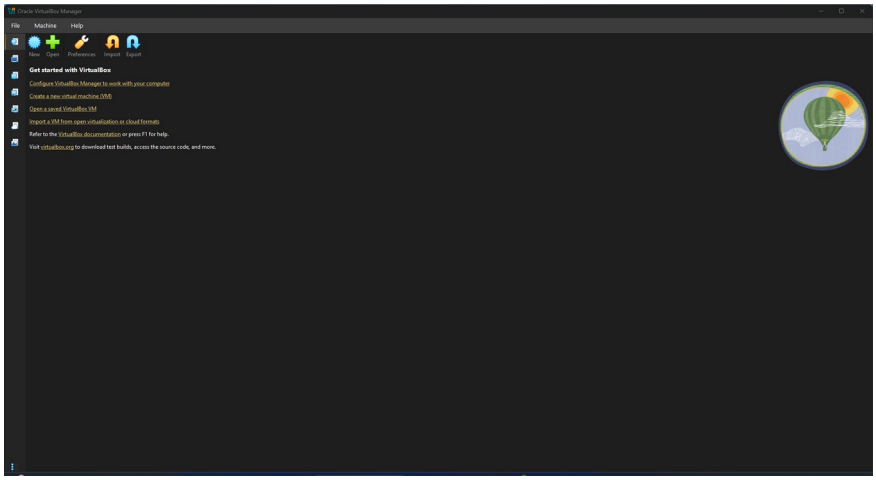
CentOS

a. Install Oracle Virtual box Manager
    i. Download Oracle Virtualbox Manager.
    ii. Go to this link https://www.virtualbox.org/wiki/Downloads
    iii. You will have to choose which host you want to install it from

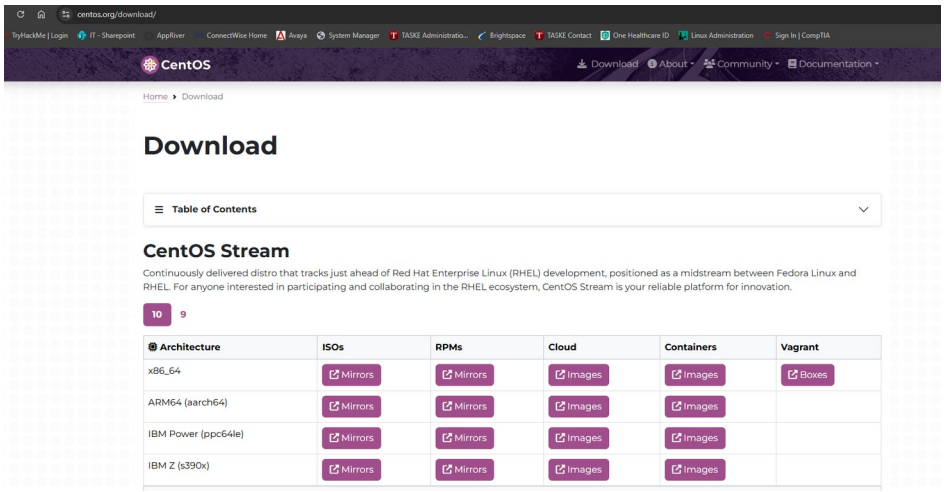


    iv. You're going to click the windows host and install virtual box
    v. Click through all the default settings and agree to terms

vi. Your screen will look like this



b. Show how to install OS
   i. Download CentOS ISO.
   ii. Go to this link https://www.centos.org/download/
   iii. You will download from version 10 - Architecture: x86_64 - ISOs

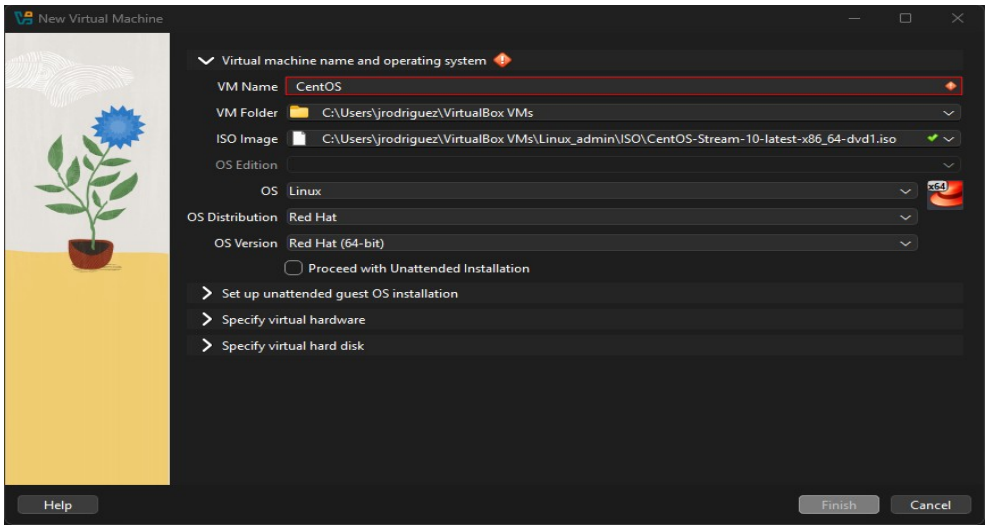


   iv. The download is going to take awhile so be patient
   v. Be sure to know the folder path of where the ISO was downloaded.

c. Add the ISO to Virtual Box manager
   i. Click the blue “new” button the top of the screen



ii. We're going to insert a name for the VM and the file and insert it into the ISO image section.



iii. Uncheck "Proceed with Unattended Installation"

iv. Go to virtual hardware to set up the RAM.

v. Go to virtual hard disk to set up the storage.

vi. Click finish and the VM should be ready to go.

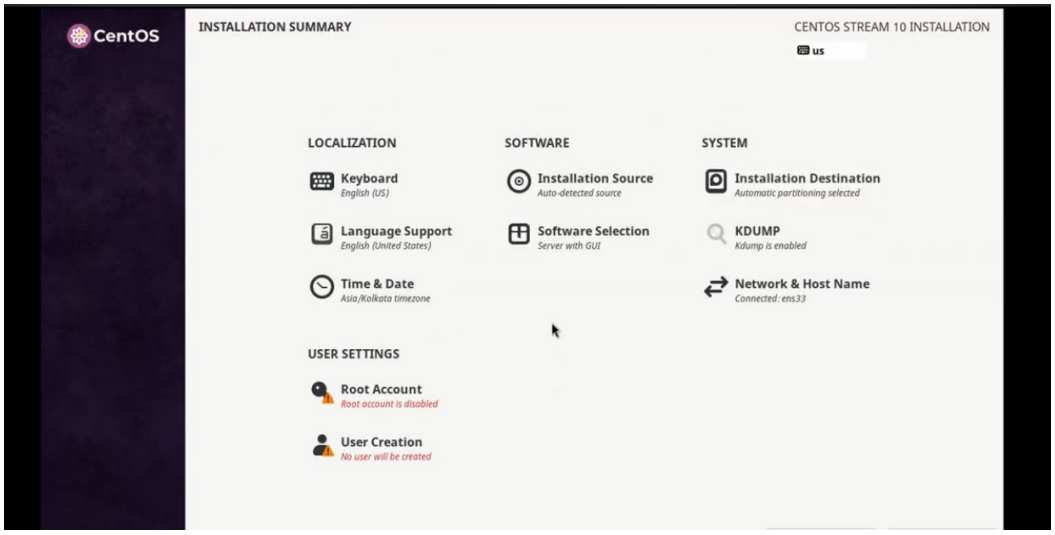
vii. Double click on the VM and the server should start running.

viii. When your server starts up select "Install CentOS Stream 10"

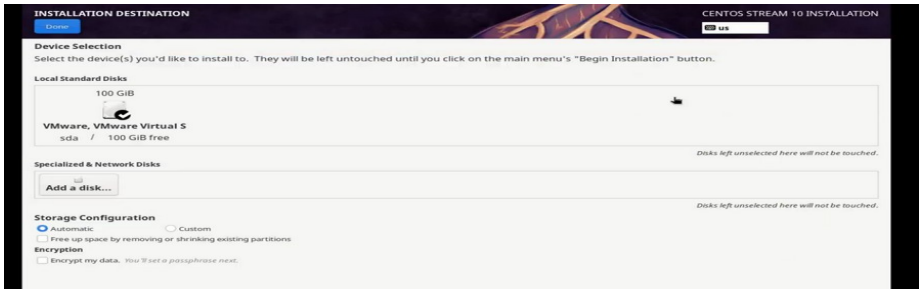


ix. When you get to the "Installation Summary" you just got to create the root account, user account select the software and

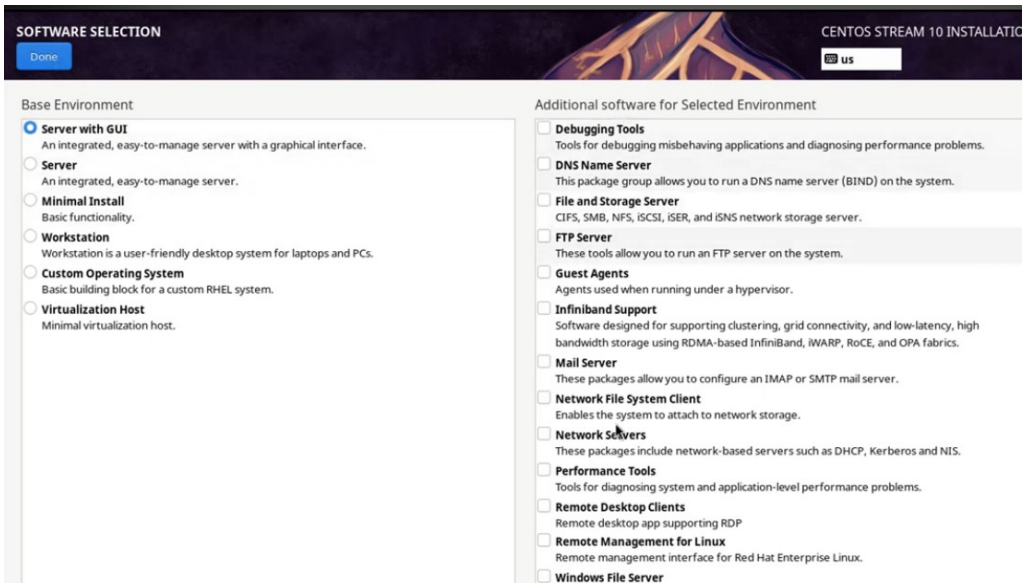
set up the Installation Destination.



x. When you open Installation Destination, it will auto select the disk all you have to do is click done.

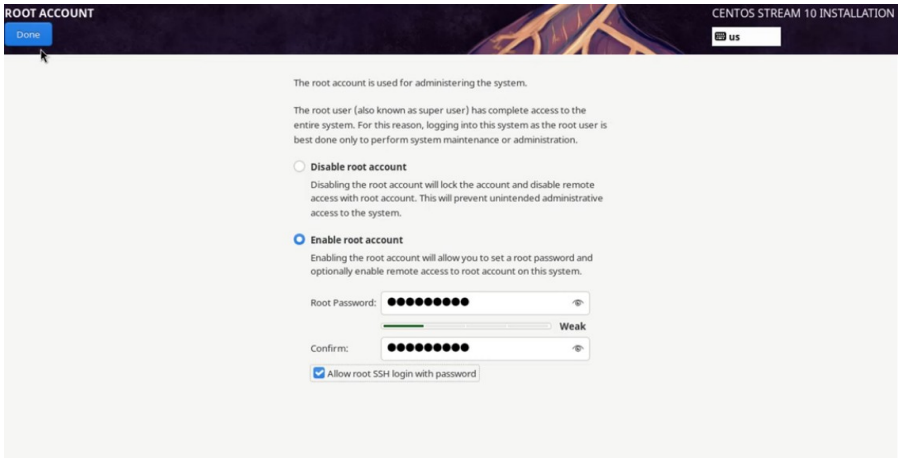


xi. Open Software Selection and select Server.



xii. Open root account, enable it make a password and check off “Allow root SSH login with password” (this will come in handy

later)

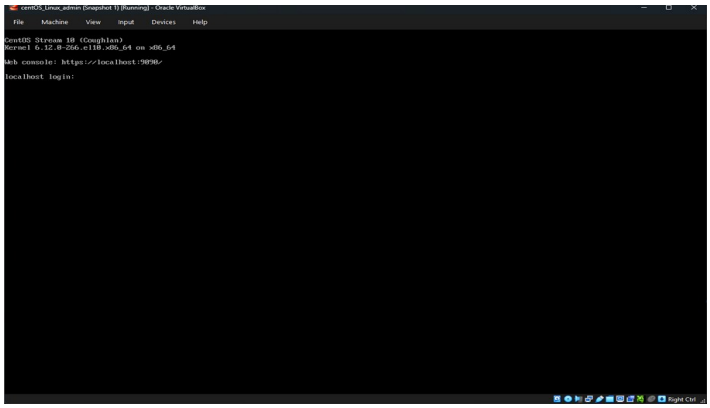


xiii. Open User Creation, fill out the full name, username, and password.

xiv. In the bottom of the Installation summary we're going to click "begin installation" and let the installation run.

xv. When the installation is done reboot the system.

xvi. The Server should now be booted as a CLI



d. How to switch the Server from a GUI to a CLI (this is if you skipped the software selection part)

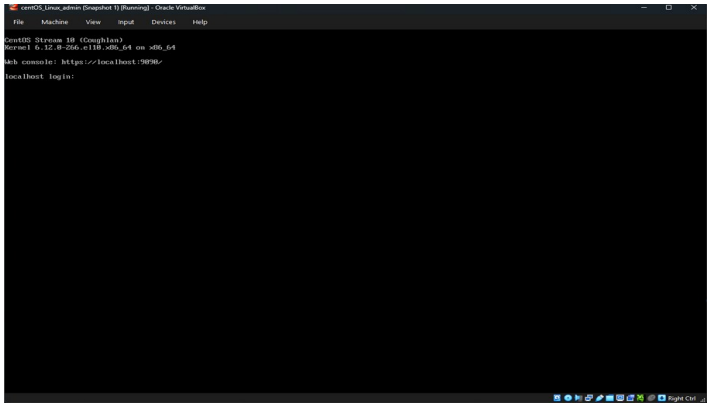
i. Open the terminal

ii. We're going to check the current default target with this command: *sudo systemctl get-default*

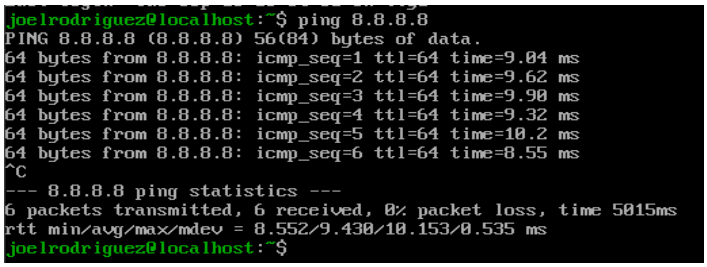
iii. Then to switch to the CLI we got to run this command: *sudo systemctl set-default multi-user.target*

iv. Reboot and confirm the appropriate interface launches: *sudo systemctl reboot*

v. Now the server should boot as CLI

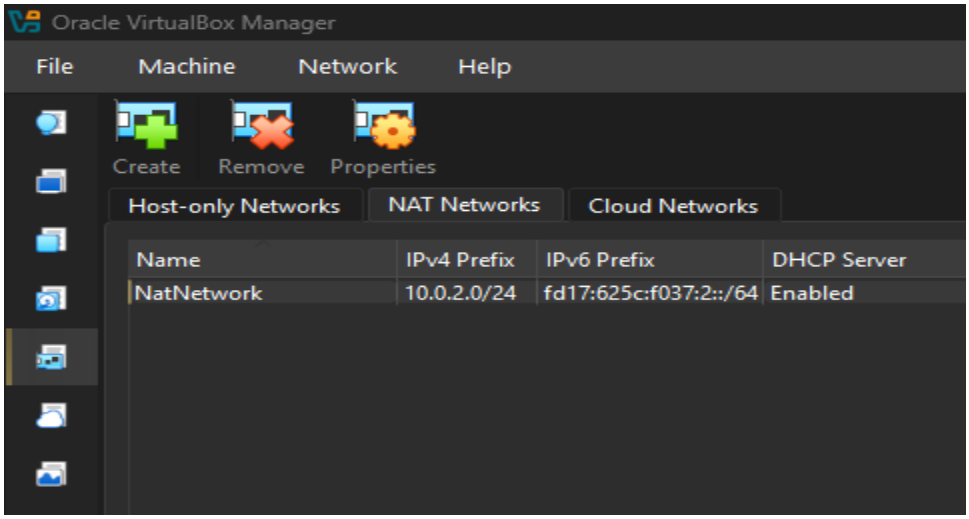


e. Show how to check internet
   i. Start up the server.
   ii. Log in with your credentials
   iii. *Ping 8.8.8.8* or *ping google.com*, if it's successful then you know you're connected to the internet.



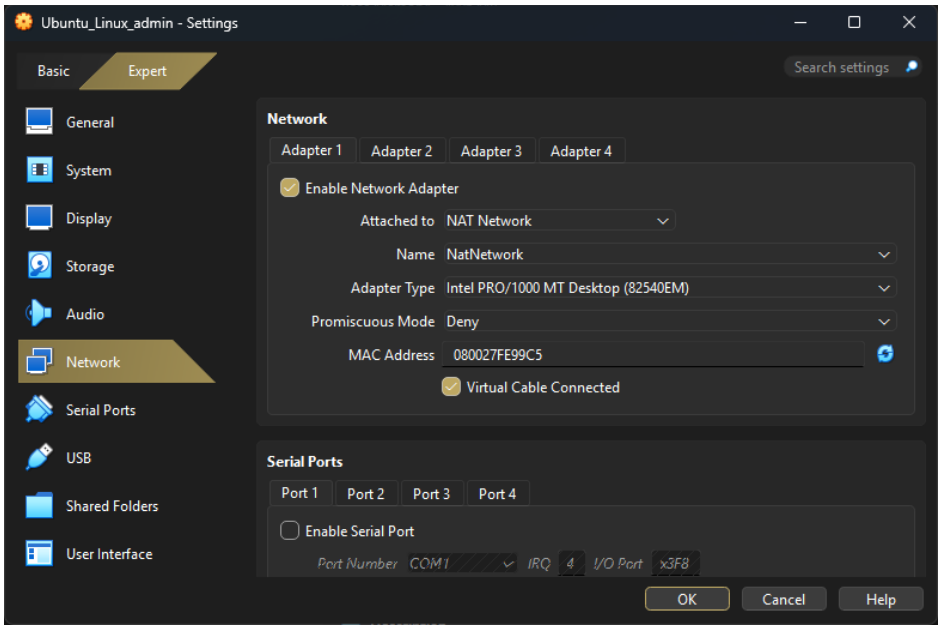
f. Show how to share file
   i. We got to configure the network to make all the VMs on the same network.
      1. Close the VM
      2. Top left corner, go to File, go to tools, open Network
      3. Open the "NAT Networks" tab
      4. Create a new network

5. NatNetwork has been created



6. Now we're going to click on CentOS VM once so we can highlight it.

7. Then go to Settings, open the network tab, Switch the "Attached to" to NATNetwork and the Name will autofill to the network we just created.



ii. Now we're going to start up CentOS

iii. Create a test file and leave a message so we can make sure everything transfers.

Command: *echo "Hello from my linux admin class!" > lab_test.txt*

```
joelrodriguez@localhost:~$ echo "Hello from my linux admin class!" > lab_test.txt
```

iv. Confirm the file was created.

Command: *ls -l*

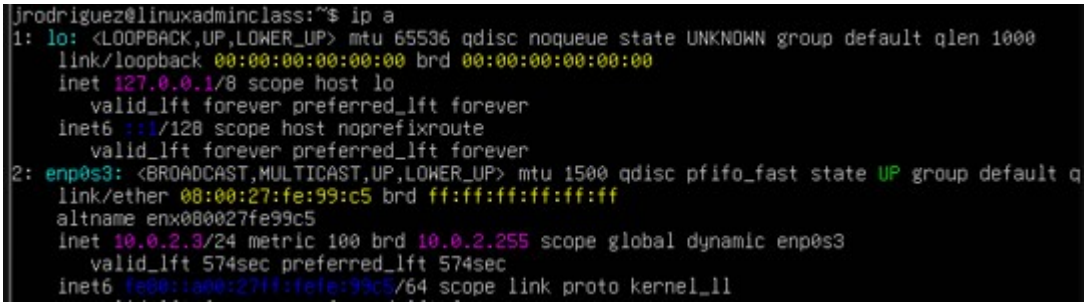
```
joelrodriguez@localhost:~$ ls -l
total 4
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Desktop
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Documents
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Downloads
-rw-r--r--. 1 joelrodriguez joelrodriguez 33 Sep 10 13:23 lab_test.txt
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Music
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Pictures
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Public
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Templates
drwxr-xr-x. 2 joelrodriguez joelrodriguez  6 Sep  9 15:11 Videos
```

v. Confirm the directory of the file

Command: pwd

```
joelrodriguez@localhost:~$ pwd
/home/joelrodriguez
```

vi. Confirm the ip address of the VM we're going to transfer the file to. Command on Ubuntu VM: *ip a*

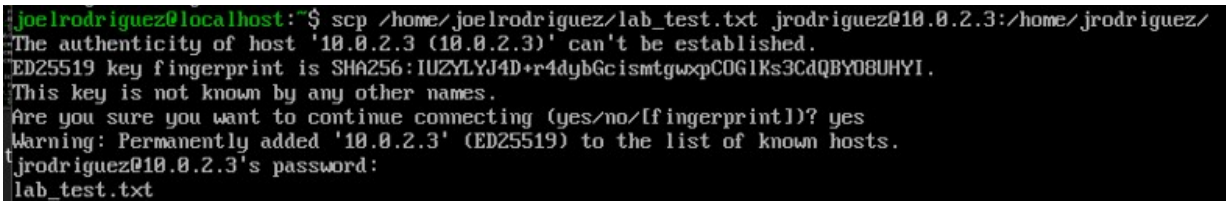


vii. Confirm the directory. Command on Ubuntu VM: *pwd*

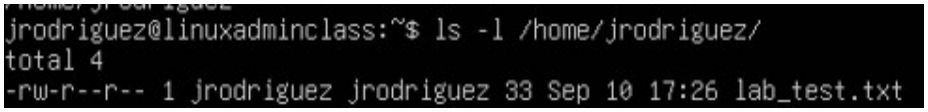


viii. Now that we know the ip address of the ubuntu vm and the directory we're sending it to, it's time to send the file over with SCP.

Command: *scp /home/joelrodriguez/lab_test.txt jrodriguez@10.0.2.3:/home/jrodriguez/*



ix. Let's check if the file made it to the Ubuntu VM
command: *ls -l /home/jrodriguez/*

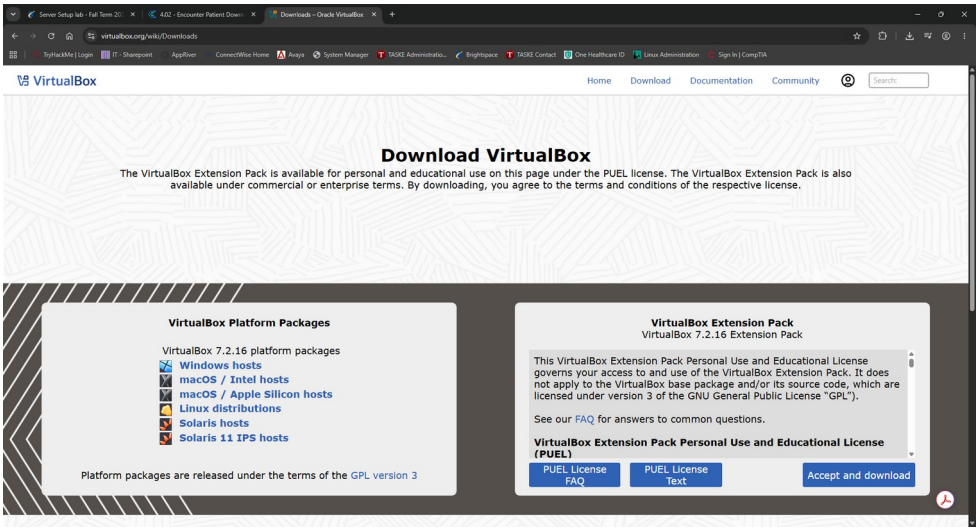


x. Confirm if the message transferred
command: *cat /home/jrodriguez/lab_test.txt*

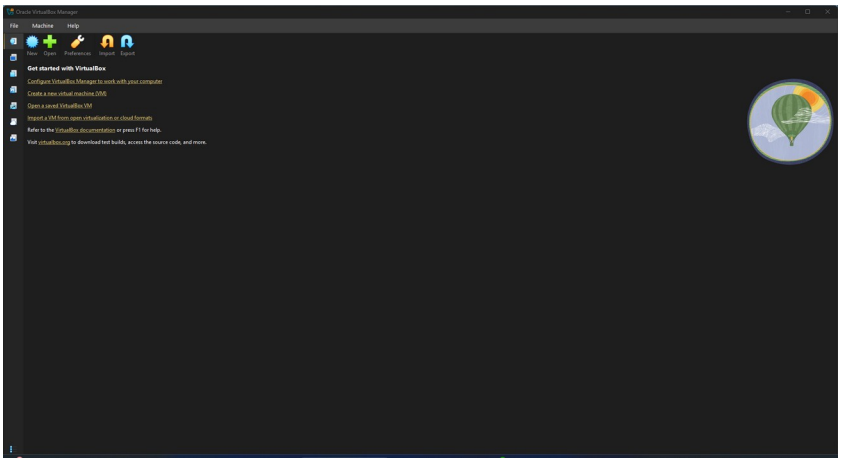


Ubuntu

a. Install Oracle Virtual box Manager
   i. Download Oracle Virtualbox Manager.
   ii. Go to this link https://www.virtualbox.org/wiki/Downloads
   iii. You will have to choose which host you want to install it from

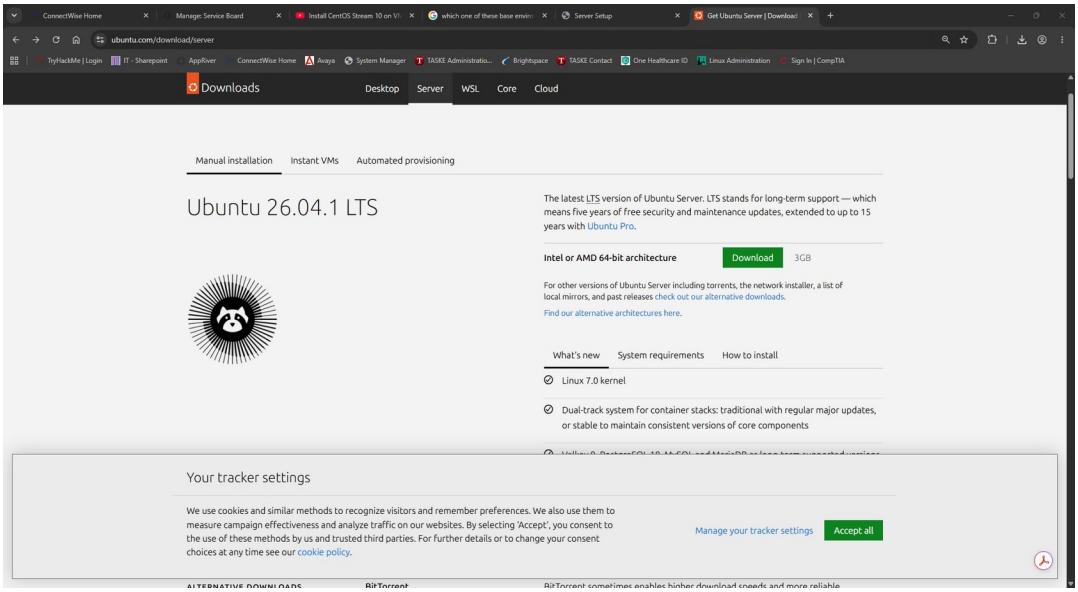


   iv. You're going to click the windows host and install virtual box
   v. Click through all the default settings and agree to terms
   vi. Your screen will look like this



b. Show how to install OS
   i. Download Ubuntu Server ISO

ii. Go to this link: https://ubuntu.com/download/server
iii. Click the green download button and the ISO will start downloading

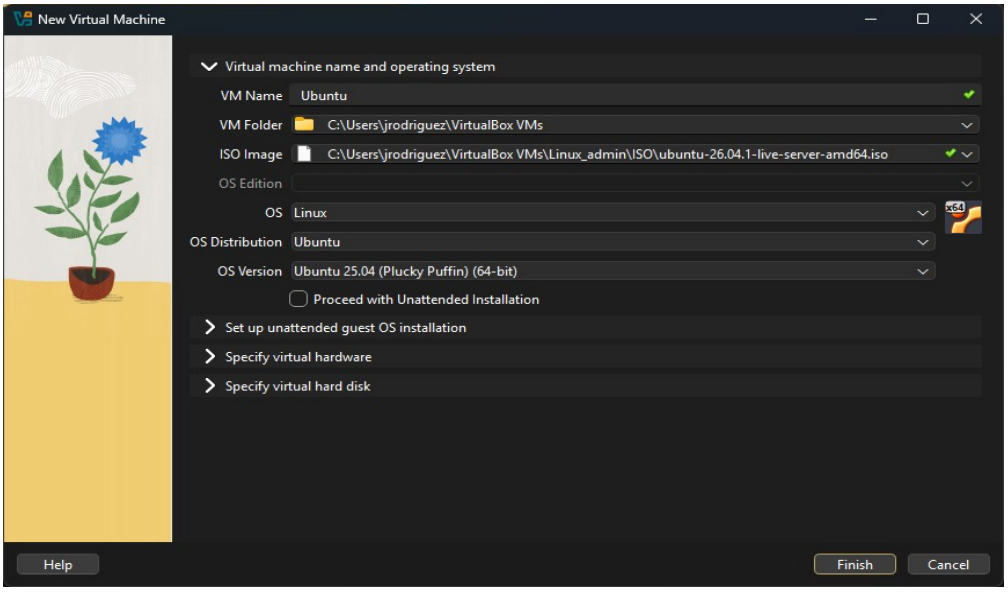


iv. Remember the path of the ISO

c. Add the ISO to Virtual Box manager
i. Click the blue “new” button the top of the screen



ii. We’re going to insert a name for the VM and the file and insert it into the ISO image section.



iii. Uncheck “Proceed with Unattended Installation”

iv. Go to virtual hardware to set up the RAM

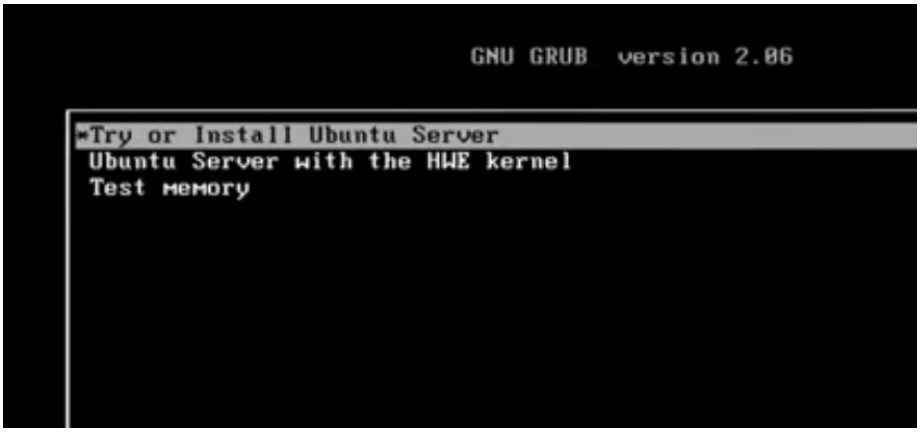
v. Go to virtual hardware to set up the storage

vi. Click finish and the VM should be ready to go

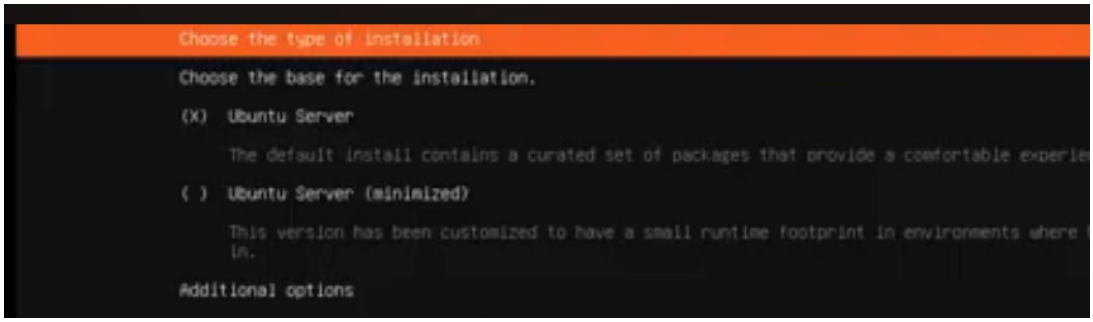
vii. Double click on the VM and the server should start running

viii. When your server starts up select “Try or install Ubuntu Server”

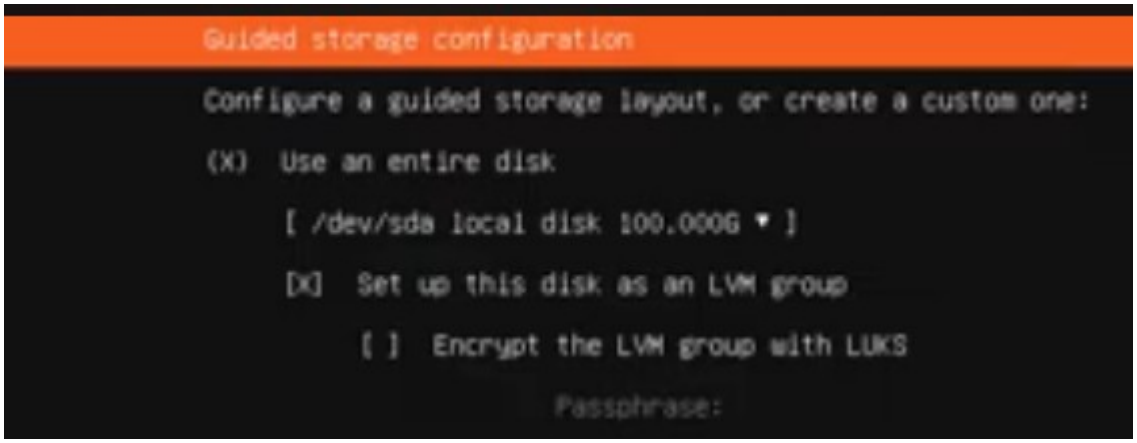
ix. Select through all the default options



x.



xi.



xii. Fill in your name, server name, username and password



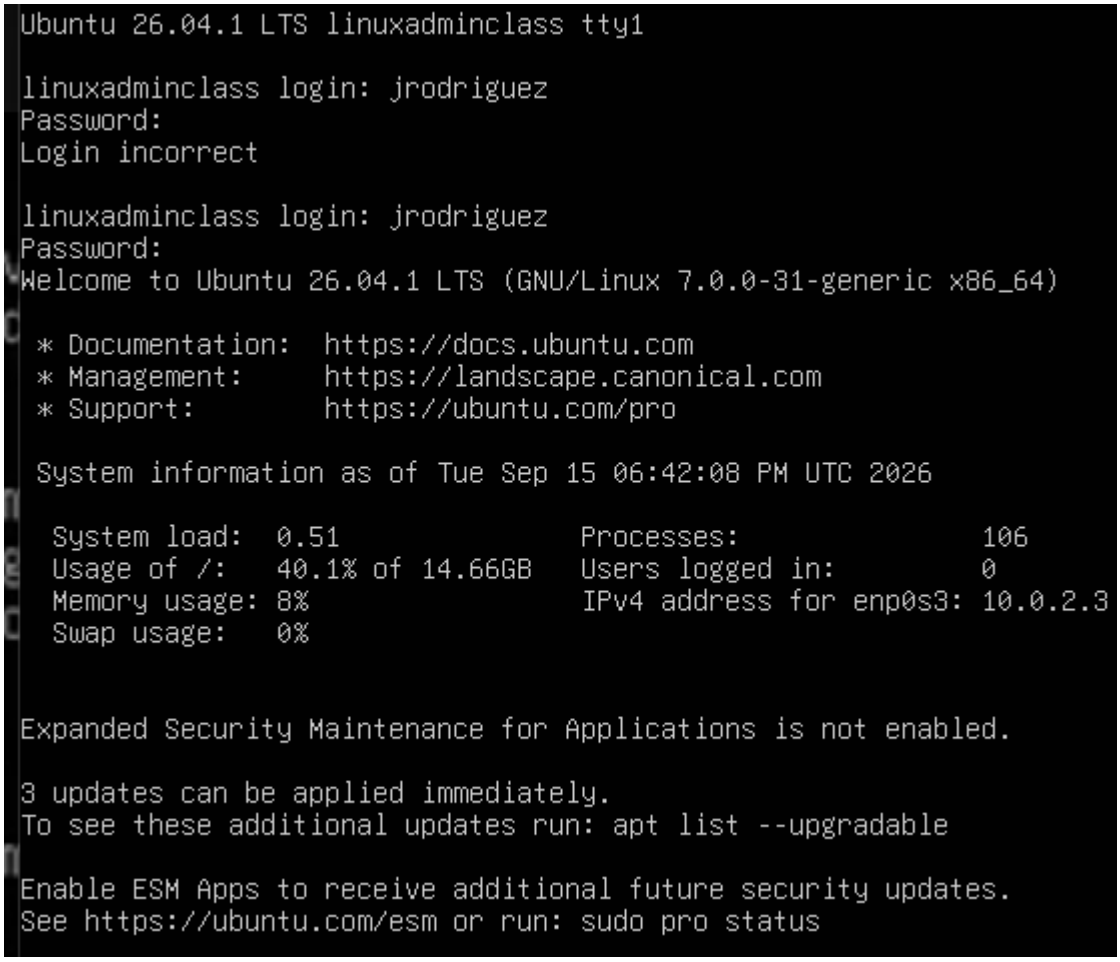
xiii. Skip upgrading to ubuntu pro

xiv. Check off “install openssh server”

xv. Click through until the system starts installing

xvi. Reboot when the install is done

xvii.



d. Show how to check internet
    i. Start up the server
    ii. Log in with your credentials

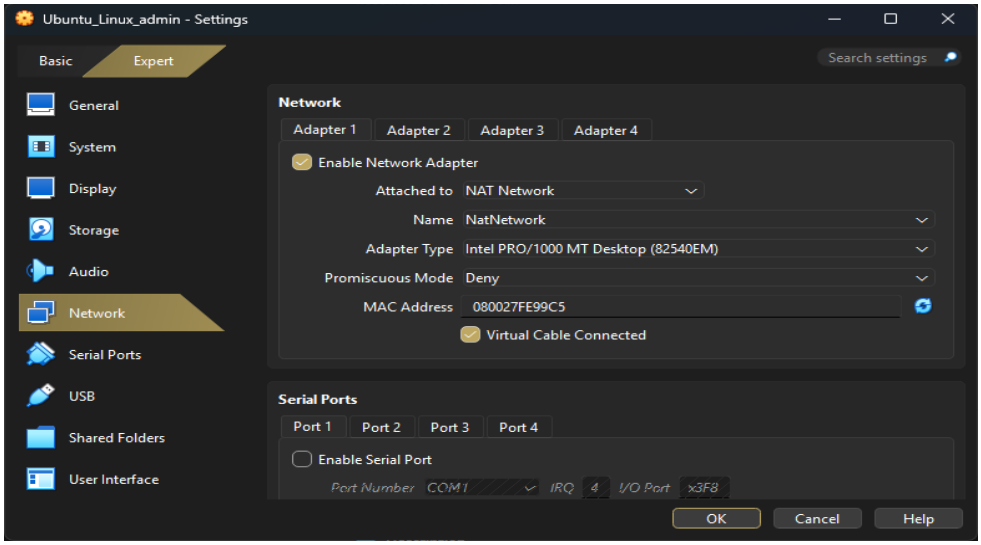
iii. *Ping 8.8.8.8* or *ping google.com*, if it's successful then you know you're connected to the internet.

```
jrodriguez@linuxadminclass:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=8.95 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=8.97 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=9.17 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=11.0 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=64 time=9.62 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 8.945/9.542/11.004/0.770 ms
jrodriguez@linuxadminclass:~$
```

e. Show how to share file

i. We got to configure the network to make all the VMs on the same network.

1. Close the VM
2. We're going to click on Ubuntu Server VM once so we can highlight it.
3. Then go to Settings, open the network tab, Switch the "Attached to" to NATNetwork and the Name will autofill to the network we just created.



ii. We're going to edit the test file from earlier and send it back to the CentOS computer just to make sure the transfer works both ways.

iii. Edit the text file. Command: *nano lab_test.txt*



iv. Now that we know the ip address of the centOS vm and the directory we're sending it to, it's time to send the file over with SCP.

Command: *scp /home/jrodriguez/lab_test.txt joelrodriguez@10.0.2.15:/home/jrodriguez*

```
jrodriguez@linuxadminclass:~$ scp /home/jrodriguez/lab_test.txt joelrodriguez@10.0.2.15:/home/joelrodriguez/
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is: SHA256:KbYHp8Nnebl6/CEgwwjp8ny0GW7Logp/r31rJ16sRHA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.
joelrodriguez@10.0.2.15's password:
lab_test.txt
```

v. Confirm that the CentOS VM received the transferred file.
Command: *la -l* and *cat lab_test.txt*

